

27) Show that $(p \rightarrow r) \wedge (q \rightarrow r)$ and $(p \vee q) \rightarrow r$ are logically equivalent.

Soln: Let $(p \rightarrow r) \wedge (q \rightarrow r)$ be True. By defn. of AND, $(p \rightarrow r)$ is True and $(q \rightarrow r)$ is True. From the defn. of conditional, the following cases arise: ① r is True. In this situation, the truth values of p and q does not matter. ② r is False. Since both $(p \rightarrow r), (q \rightarrow r)$ is True, by the defn. of conditional, p, q are both False.

Let $(p \vee q) \rightarrow r$ be True. By the defn. of conditional, the following cases arise: ① r is True. In this situation, the truth values of $(p \vee q)$ does not matter and indirectly, the truth value of p, q does not matter. ② r is False. $\therefore$ From defn of conditional, $(p \vee q)$ is False. By defn. of OR, both p, q are False.

$\therefore (p \rightarrow r) \wedge (q \rightarrow r)$ and $(p \vee q) \rightarrow r$ are logically equivalent $\square$

28) Show that $(p \rightarrow q) \vee (p \rightarrow r)$ and $p \rightarrow (q \vee r)$ are logically equivalent.

Soln: Let $(p \rightarrow q) \vee (p \rightarrow r)$ be False. By defn. of OR, both $(p \rightarrow q)$ and $(p \rightarrow r)$ is False. By defn. of conditional, p is True but q, r are both False.

Let $p \rightarrow (q \vee r)$ be False. By defn. of conditional, p is True and $q \vee r$ is False. By defn. of OR, both q, r are False.

$\therefore (p \rightarrow q) \vee (p \rightarrow r)$ and $p \rightarrow (q \vee r)$ are logically equivalent $\square$

29) Show that $(p \rightarrow r) \vee (q \rightarrow r)$ and $(p \wedge q) \rightarrow r$ are logically equivalent.

Soln: Let $(p \rightarrow r) \vee (q \rightarrow r)$ be False. By defn. of OR, both $(p \rightarrow r), (q \rightarrow r)$ are False. By defn. of conditional, p is True, q is True but r is False.

Let $(p \wedge q) \rightarrow r$ be False. By defn. of conditional, $(p \wedge q)$ is True but r is False. By defn. of AND, both p, q are True.

$\therefore (p \rightarrow r) \vee (q \rightarrow r)$ and $(p \wedge q) \rightarrow r$ are logically equivalent $\square$